Diabetes Prediction Model

Project Overview

Machine learning model developed to predict diabetes risk using patient health attributes.

Business Problem

Early detection of diabetes can significantly improve treatment outcomes.

Dataset & Features

Typical features include glucose, BMI, age, insulin levels, pregnancies, blood pressure, and skin thickness.

Data Preprocessing

Missing value treatment, normalization, feature selection, train-test split, and exploratory analysis.

Model Development

Random Forest algorithm used due to robustness and ability to handle non-linear relationships.

Evaluation Metrics

Accuracy, Precision, Recall, F1-Score, Confusion Matrix, and ROC-AUC analysis.

Challenges & Solutions

Class imbalance handling, overfitting prevention, feature importance analysis.

Deployment Possibilities

Web application using Flask or Streamlit with real-time prediction support.

Interview Questions

Why Random Forest, feature selection methods, bias-variance tradeoff, evaluation metrics, and deployment strategies.